

TF-Number Experiment 2

This experiment holds the input at 3,000 highly variable genes and varies the requested TF count. The bipartite-retained count is measured by `limit-tfs.py`; the causal-graph count is measured after top-15 TF subsetting by `grn_creation.py`.

$$\widehat{M}(x) = -4.9099x + 16093.5 \text{ MiB}, \quad R^2 = 0.9954 \quad (1)$$

$$\widehat{t}(x) = -0.06489x + 239.09 \text{ ms/step}, \quad R^2 = 0.9761 \quad (2)$$

The fits are ordinary least-squares fits over the completed runs and use requested TFs as x .

0.1 Requested TFs versus Target Genes

Target-gene count is not a better predictor in this experiment: its linear fits have $R^2 = 0.9954$ for memory and $R^2 = 0.9761$ for time, identical to the requested-TF fits. This is expected because the runs use a fixed 3,000-gene input and satisfy $\text{targets} = 3,000 - \text{requested TFs}$ (apart from graph genes excluded for lack of regulators). The equivalent target-based fits are:

$$\widehat{M}(y) = 4.9099y + 1363.9 \text{ MiB}, \quad R^2 = 0.9954 \quad (3)$$

$$\widehat{t}(y) = 0.06489y + 44.43 \text{ ms/step}, \quad R^2 = 0.9761 \quad (4)$$

Table 1: Requested and effective TF counts, plus final causal-graph size.

Requested	Bipartite retained	Graph TFs	Targets	Genes	Possible edges	Imposed edges	Density
100	100	100	2,900	3,000	290,000	41,557	14.3%
200	200	200	2,800	3,000	560,000	41,610	7.4%
300	300	300	2,700	3,000	810,000	40,362	5.0%
400	400	400	2,600	3,000	1,040,000	38,934	3.7%
500	500	500	2,500	3,000	1,250,000	37,469	3.0%
600	600	600	2,400	3,000	1,440,000	35,986	2.5%
700	700	700	2,300	3,000	1,610,000	34,499	2.1%
800	800	800	2,200	3,000	1,760,000	33,000	1.9%
900	900	900	2,100	3,000	1,890,000	31,500	1.7%
1,000	1,000	1,000	2,000	3,000	2,000,000	30,000	1.5%
1,100	1,100	1,100	1,900	3,000	2,090,000	28,500	1.4%
1,200	1,200	1,200	1,800	3,000	2,160,000	27,000	1.2%
1,300	1,300	1,300	1,700	3,000	2,210,000	25,500	1.2%
1,400	1,400	1,399	1,600	2,999	2,238,400	24,000	1.1%
1,500	1,500	1,496	1,500	2,996	2,244,000	22,500	1.0%
1,600	1,600	1,586	1,400	2,986	2,220,400	21,000	0.9%
1,700	1,699	1,668	1,300	2,968	2,168,400	19,500	0.9%
1,800	1,798	1,739	1,200	2,939	2,086,800	18,000	0.9%
1,900	1,894	1,803	1,100	2,903	1,983,300	16,500	0.8%
2,000	1,988	1,844	1,000	2,844	1,844,000	15,000	0.8%
2,100	2,076	1,872	900	2,772	1,684,800	13,500	0.8%
2,200	2,163	1,873	800	2,673	1,498,400	12,000	0.8%
2,300	2,244	1,834	700	2,534	1,283,800	10,500	0.8%
2,400	2,311	1,782	600	2,382	1,069,200	9,000	0.8%
2,500	2,354	1,699	500	2,199	849,500	7,500	0.9%
2,600	2,351	1,577	400	1,977	630,800	6,000	1.0%
2,700	2,318	1,406	300	1,706	421,800	4,500	1.1%
2,800	2,178	1,155	200	1,355	231,000	3,000	1.3%
2,900	1,853	792	100	892	79,200	1,500	1.9%

Table 2: Measured causal-generator time and memory requirements.

Requested TFs	Target genes	Avg step time (ms)	Peak memory (MiB)
100	2,900	226	14,545
200	2,800	236	14,843
300	2,700	217	14,829
400	2,600	211	14,365
500	2,500	219	13,541
600	2,400	212	13,049
700	2,300	192	13,059
800	2,200	184	12,393
900	2,100	178	11,881
1,000	2,000	157	11,125
1,100	1,900	164	11,041
1,200	1,800	173	10,371
1,300	1,700	150	9,869
1,400	1,600	149	9,029
1,500	1,500	151	8,863
1,600	1,400	131	8,373
1,700	1,300	134	7,865
1,800	1,200	107	7,365
1,900	1,100	116	6,857
2,000	1,000	114	6,371
2,100	900	109	5,869
2,200	800	81	5,043
2,300	700	78	4,465
2,400	600	88	4,295
2,500	500	67	3,787
2,600	400	72	3,419
2,700	300	65	2,885
2,800	200	69	1,899
2,900	100	61	1,837

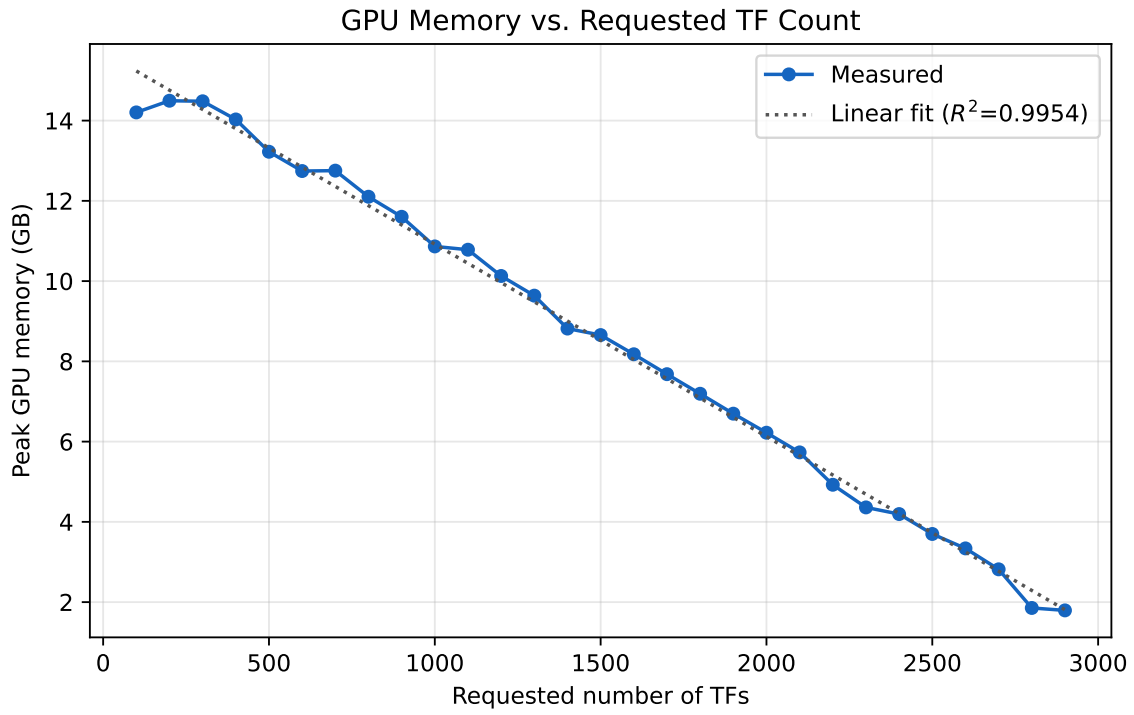


Figure 1: Peak GPU memory requirement versus requested TF count.

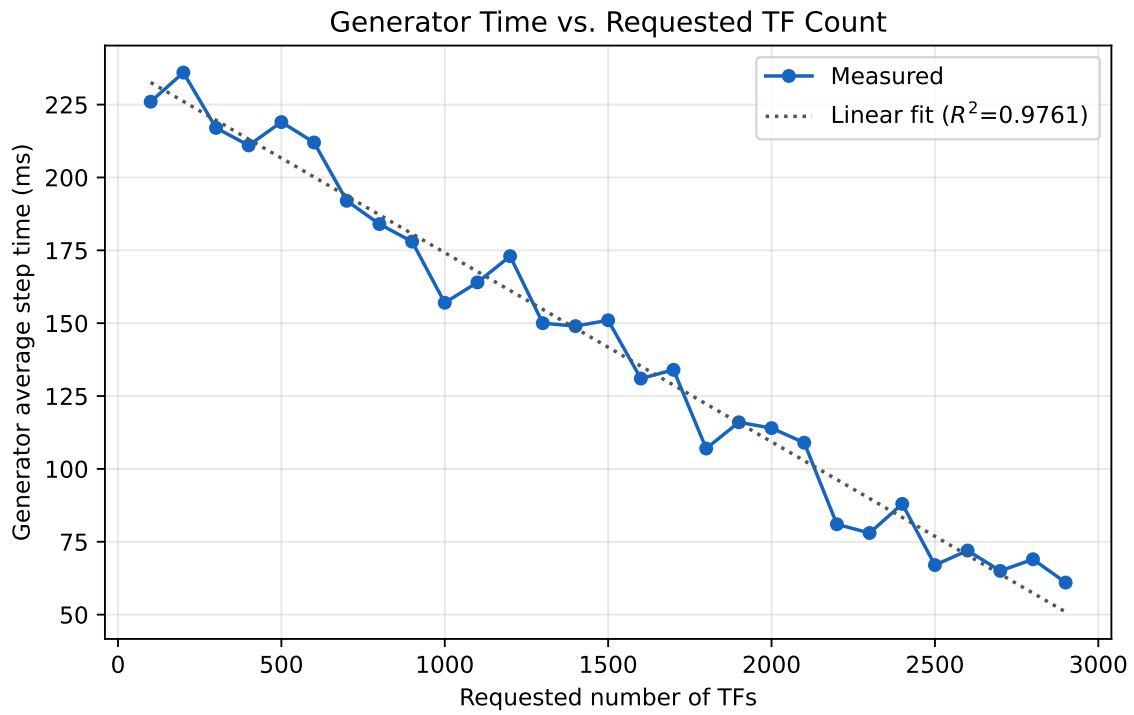


Figure 2: Causal-generator average step time versus requested TF count.